

Mutual Fund Performance, Risk & Investor Analytics Platform

Data Analyst Internship Project

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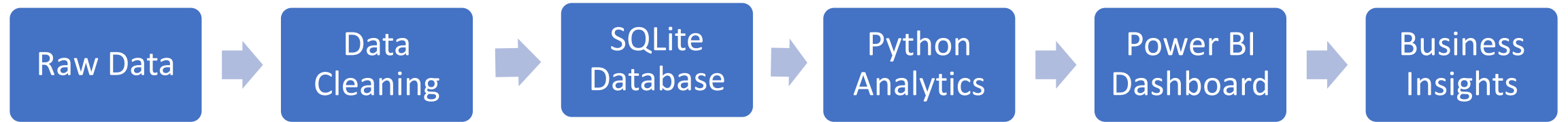
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Python | SQLite | Power BI

Problem Statement

- • Large volume of mutual fund data
 - • Difficulty tracking performance
 - • Need for risk-adjusted evaluation
 - • Limited investor retention insights
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- Objective:
 - Build an end-to-end analytics platform for performance,
 - risk, and investor behavior analysis.

Project Architecture

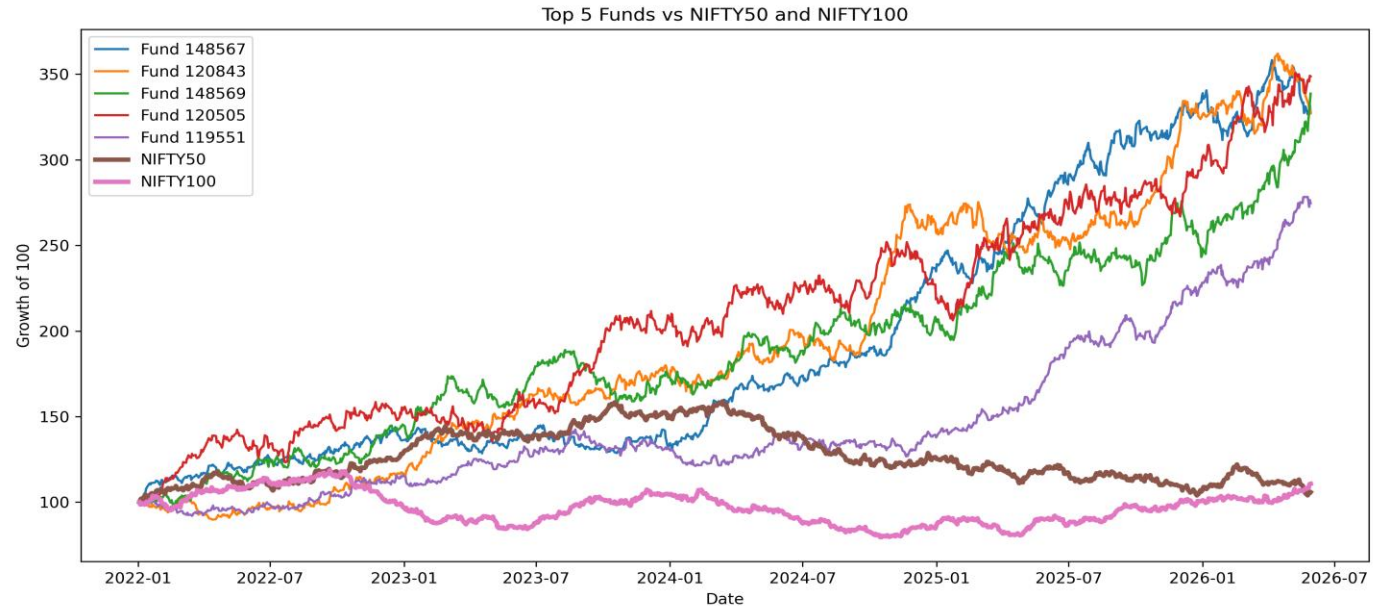
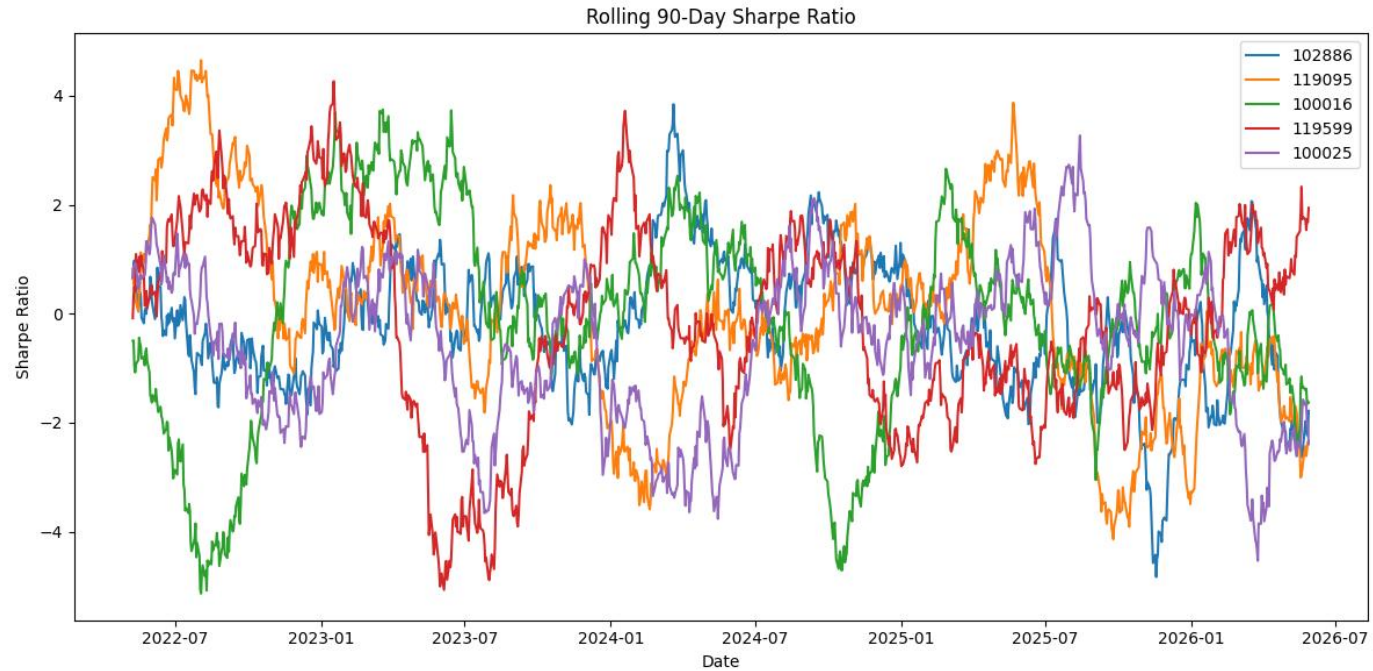


Dataset Overview

Dataset	Purpose
Fund Master	Fund Information
NAV History	Performance Tracking
Transactions	Investor Analysis
Benchmark Data	Market Comparison
Holdings	Portfolio Analysis
SIP Inflows	Investment Trends

Performance Analytics

- Key Metrics Calculated
 - Alpha
 - Measures excess return over benchmark
 - Beta
 - Measures market sensitivity
 - Sharpe Ratio
 - Measures risk-adjusted return
- Benchmark Comparison
 - Compares fund performance against market indices



Risk Analytics

- Risk Measures Implemented
 - Value at Risk (VaR)
 - Conditional VaR (CVaR)
 - HHI Concentration Analysis
- Outputs:
 - `var_cvar_report.csv`
 - `hhi_concentration_report.csv`
- Goal:
 - Measure downside risk and portfolio concentration.

Investor Analytics

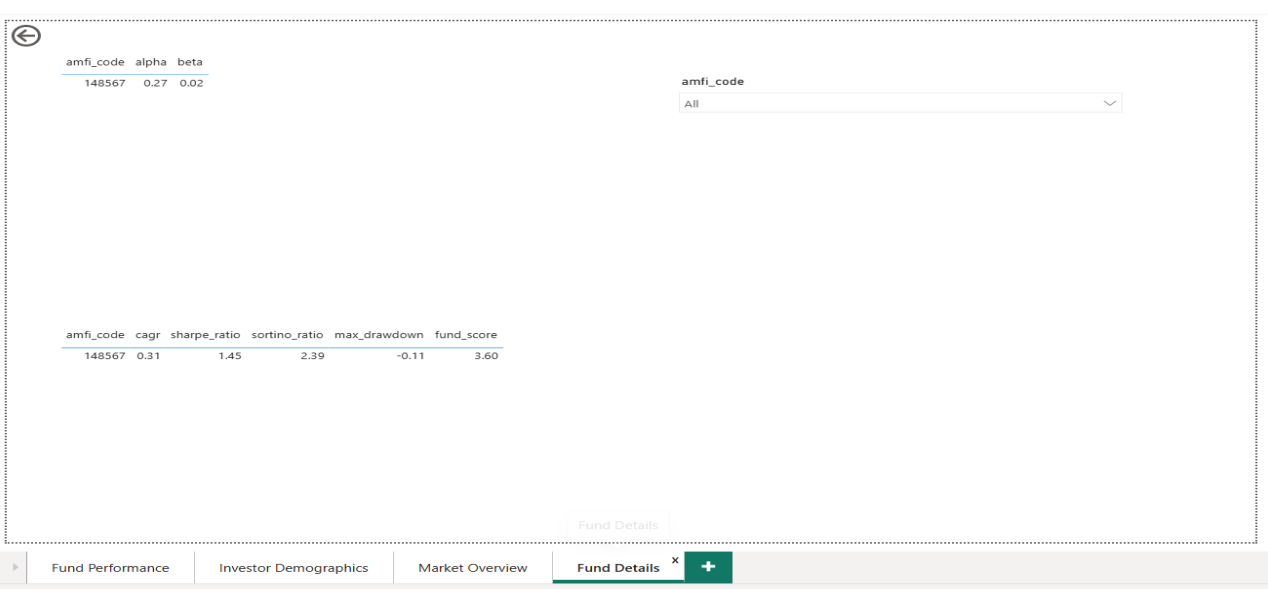
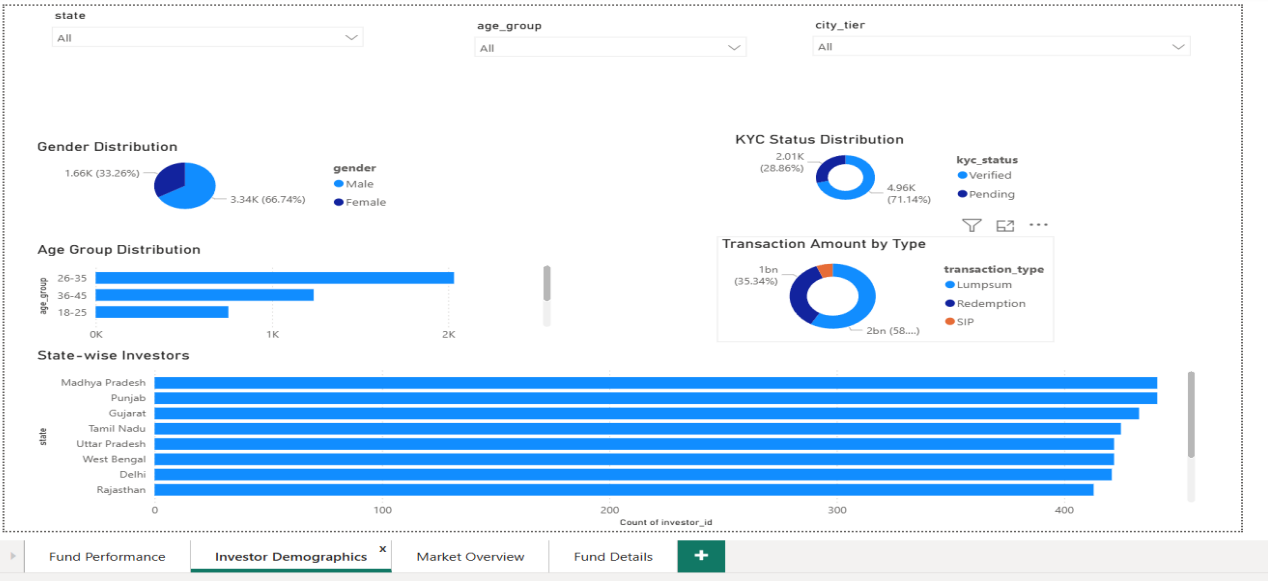
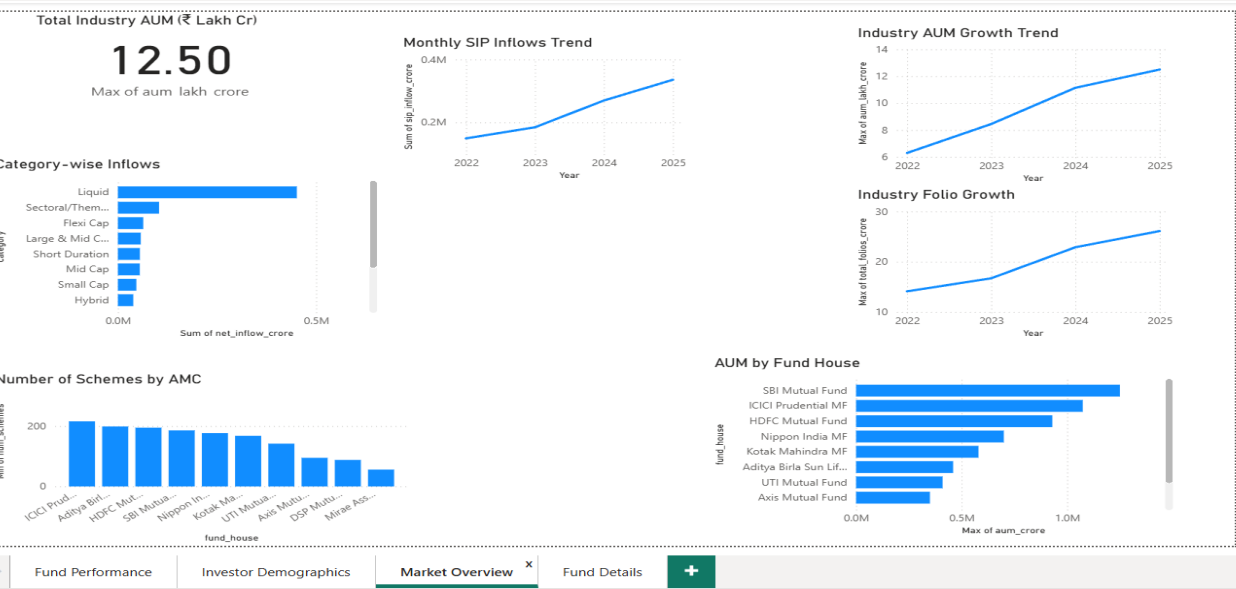
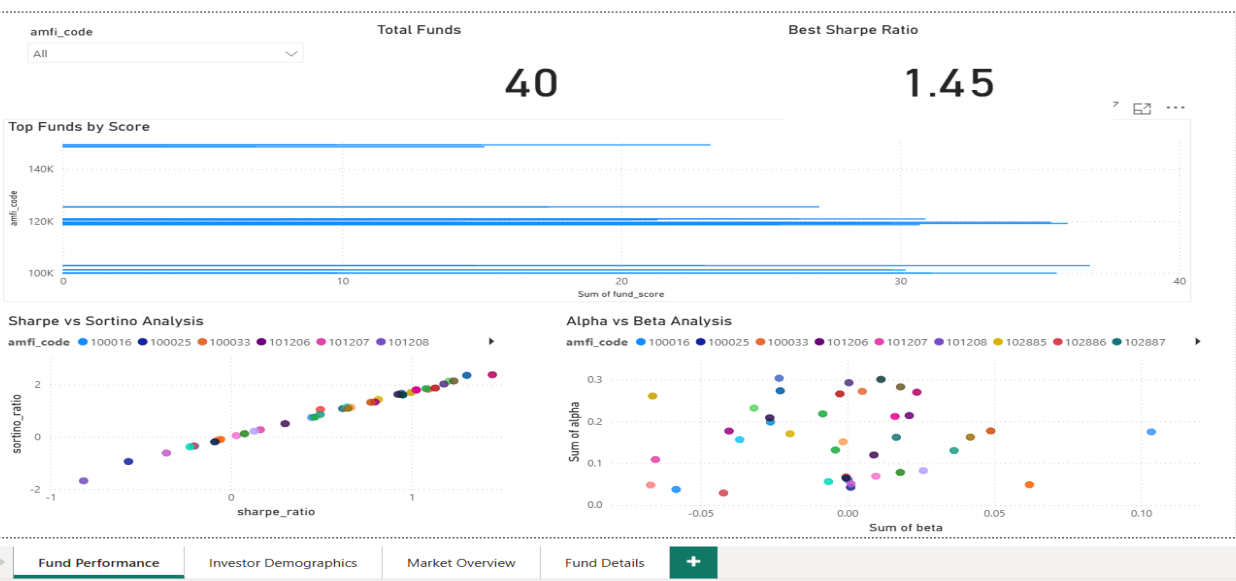
- Analyses Performed
 - Cohort Analysis
 - Groups investors based on their first investment period.
 - SIP Continuity Analysis
 - Measures consistency of SIP investments over time.
- Key Insights
 - Investor activity is highest during initial months.
 - Retention decreases gradually over time.
 - Long-term investors contribute significantly to fund stability.
 - SIP continuity helps identify loyal investors.

Business Insight:
Investor retention and SIP continuity are critical indicators of long-term mutual fund growth.

Fund Recommendation Engine

- Developed using Python
- Input:
 - • Fund Scorecard
 - • Risk Metrics
 - • Sharpe Ratio
- Output:
 - Top-ranked mutual funds based on
 - risk-adjusted performance.

Power BI Dashboard



- Business Impact:
- Interactive fund performance monitoring
 - Investor segmentation and demographic analysis
 - Market trend tracking and decision support

Project Outcomes & Conclusion

- Key Achievements
 - ✓ Data Cleaning & Validation
 - ✓ SQLite Database Development
 - ✓ Performance Analytics
 - ✓ Risk Analytics (Alpha, Beta, VaR, CVaR)
 - ✓ Investor Analytics
 - ✓ Recommendation Engine
 - ✓ Power BI Dashboard
- Business Impact
 - • Better fund performance evaluation
 - • Improved risk assessment
 - • Investor retention insights
 - • Data-driven decision making
 - • Interactive business reporting

Mutual Fund Performance, Risk & Investor Analytics Platform

Thank You

Questions?